

Nick Tusay

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Physics-Astronomy Bldg

Seattle, WA 98195

Research Focus:

Radio technosignature search methods, observations and data analysis.

Exoplanet transmission spectroscopy with JWST and ground-based optical observatories.

Education:

Ph.D. in Astronomy & Astrophysics and Astrobiology — August 2025

Thesis: Searching for Signs of Life and Habitable Worlds

Penn State University, State College, PA

M.S. in Astronomy & Astrophysics — May 2023

Penn State University, State College, PA

B.S. in Physics — May 2020

The College of New Jersey, Ewing, NJ

B.S. in Mechanical Engineering — May 2004

Rutgers University, New Brunswick, NJ



Research Experience:

Postdoctoral Research at University of Washington

Advisor: Prof. James Davenport

Mid-IR Spectroscopy of BD+05 4868 Ab with JWST

October 2025 - present

- Co-PI for JWST GO program 8739, MIRI LRS transmission spectroscopy of BD+05 4868 Ab
- Contributed to and submitted successful JWST cycle 4 proposal. Awarded \$271,965 for 35 hrs of science time
- Simulated JWST MIRI data using PandExo and built a linear regression code in Python to fit a suite of geophysical composition models
- Reduced JWST MIRI LRS data from MAST using the Eureka! pipeline
- Fit transit models to the lightcurve and extracted a transmission spectrum from model depths in each channel
- Constructed geophysical models of mineralogical compounds based on mie scattering theory

VGRF Research at IPAC/Caltech

Advisor: Dr. Aurora Kesseli

High-res Transmission Spectroscopy of WASP-82b with ESPRESSO and KPF February 2025 - present

- Assisted with taking observations of WASP-82b using KPF on Keck
- Used petitRADTRANS to produce atmospheric templates for cross-correlation with the ESPRESSO and KPF spectra
- Cleaned and reduced high-resolution ESPRESSO and KPF data to cross-correlate with atomic line templates and produce CCFs
- Validated our retrieval pipeline by injecting atmospheric templates into our data and recovering atomic lines in the CCFs
- Obtained a projected obliquity measurement from the ESPRESSO and KPF RV measurements using an MCMC from RMfit
- Used the RM Reloaded technique to measure the RV centroid shifts during the transit of WASP-82b
- Created a forward model of the star and planet system to simulate and parameterize the RMR CCF, including stellar inclination, differential rotation, and convective blueshift center-to-limb variations
- Created and ran an MCMC to determine the best fit parameters that produce a model most consistent with the RMR CCF data
- Determined the true obliquity of WASP-82b

Doctoral Research at Penn State University

Advisor: Prof. Jason Wright

Mid-IR Spectroscopy of K2-22b with JWST

August 2022 - July 2025

- Science PI for JWST GO program 3315, MIRI LRS transmission spectroscopy of K2-22b
- Led and submitted successful JWST cycle 2 proposal. Awarded \$261,526 for 18 hrs of science time
- Simulated JWST MIRI data using PandExo and built a linear regression code in Python to fit a suite of geophysical composition models
- Extended model retrieval simulations to optical wavelengths, and submitted successful proposals for simultaneous optical observations to LCO, WIYN, and CHEOPS

- Coordinated simultaneous optical observations at LCO, WIYN, APO, the 88" telescope at UH, and CHEOPS
- Reduced JWST MIRI LRS data from MAST using the Eureka! pipeline
- Fit transit models to the lightcurve and extracted a transmission spectrum from model depths in each channel
- Constructed geophysical models of mineralogical compounds based on mie scattering theory
- PI of 3 Cycle 4 proposals following up on K2-22b and 2 additional disintegrating planet candidates

Doctoral Research at Penn State University

Advisor: Prof. Jason Wright

SETI during PPOs in TRAPPIST-1 with ATA

August 2022 - December 2024

- Used **turboSETI** to identify narrowband signals in the high frequency resolution filterbank data products from observations of TRAPPIST-1 at the Allen Telescope Array
- Built a python based pipeline to filter and plot the detected signals by leveraging beamforming to discriminate obvious sources of radio frequency interference
- Tested my algorithm by applying **setigen** to perform injection recovery of simulated signals
- Modified and employed the julia code **NbodyGradient** to determine planet-planet occultation events that may have occurred during our observations
- Plotted and examined the dynamic spectrum of thousands of candidate signals
- Wrote and published paper in the Astronomical Journal as first and corresponding author.

Doctoral Research at Penn State University

Advisor: Prof. Jason Wright

SETI at the Solar Gravitational Lens Focal Region

August 2020 - August 2022

- Assisted in carrying out observations at Green Bank Telescope
- Managed data transfer to and storage on the Penn State high performance computing cluster
- Used **turboSETI** to analyze the high frequency resolution filterbank data products
- Plotted and examined the dynamic spectrum of thousands of candidate signals
- Co-first-authored the paper, being chiefly responsible for writing the analysis and results sections

Undergraduate Research at The College of New Jersey

Advisor: Dr. Lauranne Lanz

Investigating Galactic Evolution by Measuring AGN Feedback

Fall 2019 - Spring 2020

- Learned and utilized CIAO software to count X-ray photons in the central region of NGC1266
- Used Python to model X-ray spectra and map the temperature profile of the AGN outflow

REU at Notre Dame University

Advisor: Dr. Justin Crepp

Improving Adaptive Optics Wavefront Instrumentation

May 2019 - August 2019

- Used Solidworks to render a model of our optical system
- Assisted in the assembly and alignment of multiple optical setups in an optics lab
- Helped install and use a Shack-Hartmann Wave Front Sensor in our optical setup
- Developed GUI software for the mechanical components of our system using Matlab App Designer
- Coordinated with nanofabrication engineers to produce and measure microscopic etchings to test our optical system

Undergraduate Research at The College of New Jersey

Advisor: Dr. Paul Wiita

High Resolution 3D Relativistic AGN Jet Simulations

Fall 2018 - Spring 2019

- Used **Athena** and **Athena++** code on TCNJ's computing cluster to run higher resolution simulations than previously done of AGN jets with various jet speeds and densities
- Constructed 2D and 3D images from the simulation outputs using ParaView software and compared against lower resolution runs
- Modified the **Athena++** code to include relativistic effects in the equations of state

Undergraduate Research at The College of New Jersey

Advisor: Dr. Nathan Magee

Investigating Ice Crystal Morphology in Cirrus Clouds

May 2018 - August 2018

- Modified payloads and launched weather balloons to collect ice crystals in cirrus clouds
- Analyzed weather forecasts to select optimal launch and collection times and locations
- Grew ice crystals in the lab on various substrates at various pressures, temperatures and humidity levels to simulate ice crystal formation in cirrus clouds
- Used an ellipsometer to obtain polarization data on grown crystals in the lab
- Worked in the machine shop to synthesize mechanical components for the lab and our payloads
- Operated the scanning electron microscope to obtain images of captured and lab-grown crystals

Teaching and Mentorship Experience:

High School Physics Teacher

- Hopewell Valley Central High School, Pennington, NJ September 2010 - June 2011
- Taught three sections of regular physics and one section of AP physics C
 - Attended night classes while teaching to obtain alternate route public school certification in NJ
- The Pennington School, Pennington, NJ September 2005 - June 2007
- Taught all three levels of physics: regular, honors and AP
 - Acquired a Commercial Driver's License (CDL) to help transport students
 - Lived on campus and assisted boarding students: monitoring, transporting, tutoring, etc.

Awards and Grants:

- JWST Cycle 4 GO Program 8739, Awarded \$271,965 for 34.8 hrs of science time (2025)
- 2025 ExoExplorer through NASA's Exoplanet Exploration Program
- Visiting Graduate Student Fellowship at IPAC (2025)
- JWST Cycle 2 GO Program 3315, Awarded \$261,526 for 18 hrs of science time (2023)
- National Science Foundation Graduate Research Fellowship Program (NSF GRFP) Awardee (2020)
- Distinguished Graduate Fellowship at Penn State University (2020)

Skills:

- Proficient Languages: Python, Julia
- Familiar Languages: Fortran, R, C++, Matlab
- Software: petitRADTRANS, RMfit, turboSETI, NbeamAnalysis, Eureka, AstroImageJ
- Telescopes Observing Experience: GBT, ATA, WIYN, LCO, Keck, Palomar, CHEOPS, JWST
- Other Skills: Linux, Bash, Github, Excel, LaTeX, High Performance Computing, Large data volume management (~TBs)

Publications:

- [1] Morgan Saidel et al. "The First Dedicated Survey of Atmospheric Escape from Planets Orbiting F Stars". In: *The Astrophysical Journal Letters* 171.4, 257 (Apr. 2026), p. 257. DOI: 10.3847/1538-3881/ae4e17. arXiv: 2510.05240 [astro-ph.EP].
- [2] **Tusay, Nick** et al. "A Disintegrating Rocky World Shrouded in Dust and Gas: Mid-infrared Observations of K2-22 b Using JWST". In: *The Astrophysical Journal Letters* 987.1, L6 (July 2025), p. L6. DOI: 10.3847/2041-8213/addfd0. arXiv: 2501.08301 [astro-ph.EP].
- [3] Jason T. Wright et al. "SETI in 2022". In: *Acta Astronautica* 225 (Dec. 2024), pp. 1–8. DOI: 10.1016/j.actaastro.2024.09.072. arXiv: 2410.08253 [astro-ph.IM].
- [4] **Tusay, Nick** et al. "A Radio Technosignature Search of TRAPPIST-1 with the Allen Telescope Array". In: *The Astronomical Journal* 168.6, 283 (Dec. 2024), p. 283. DOI: 10.3847/1538-3881/ad823c. arXiv: 2409.08313 [astro-ph.EP].
- [5] **Nick Tusay** and Macy J. Huston et al. "A Search for Radio Technosignatures at the Solar Gravitational Lens Targeting Alpha Centauri". In: *The Astronomical Journal* 164.3, 116 (Sept. 2022), p. 116. DOI: 10.3847/1538-3881/ac8358. arXiv: 2206.14807 [astro-ph.IM].
- [6] Nathan Magee et al. "Captured cirrus ice particles in high definition". In: *Atmospheric Chemistry & Physics* 21.9 (May 2021), pp. 7171–7185. DOI: 10.5194/acp-21-7171-2021.
- [7] Justin R. Crepp et al. "Measuring phase errors in the presence of scintillation". In: *Optics Express* 28.25 (Dec. 2020), p. 37721. DOI: 10.1364/OE.408825. arXiv: 2012.12695 [astro-ph.IM].

Conference Organizing and Committee Assignments:

- The Second Assembly of the Order of the Octopus, August 2024
 - Organizing Committee member (LOC & SOC)
- PSETI Symposium at PSU, June 2023
 - Co-chair of the Local Organizing Committee (LOC)
 - Science Organizing Committee (SOC) member
- The First PSETI Symposium at Penn State University, June 2022
 - Local Organizing Committee (LOC) member

- The First Assembly of the Order of the Octopus (virtual), August 2020
 - Organizing Committee member
- Climate and Diversity Committee, September 2020 - Present
 - Contributed in the planning and execution of activities designed to improve the climate and diversity of the Astronomy and Astrophysics department, such as Astro+ Research Showcase, Career Seminar Series, Department Beautification Project, Undergraduate Town Hall
- Graduate Program Committee, September 2021 - September 2023
 - Represented graduate student voices in discussions on improving the graduate program.
 - Actively worked with graduate students and faculty to design a better qualifying exam.
- Graduate Student Office Manager, September 2022 - September 2024
 - Managed the graduate student desk assignments and redistribution of office space with the yearly departure of newly graduated students and first-year arrivals

Conference and Workshop Presentations:

- Exoplanets 6 in Porto, Portugal, June 2026
 - Poster presentation on WASP-82b results
- Cool Stars 23 in Tokyo, Japan, June 2026
 - Oral presentation on preliminary JWST results of BD+05 4868 Ab
- Exploring Planetary Systems in the Era of Time-Domain Astronomy in Honolulu, HI, June 2026
 - Oral presentation on WASP-82b results
- AASTCS: 11 Exoplanet Atmospheres in Denver, CO, March 2026
 - Oral presentation on preliminary JWST results of BD+05 4868 Ab
- American Astronomical Society (AAS) Winter Meeting 247 in Phoenix, AZ, January 2026
 - Oral presentation of my PhD thesis
- American Astronomical Society (AAS) Summer Meeting 246 in Anchorage, AK, June 2025
 - Oral presentation on preliminary results of VGRF project on WASP-82b
- Kavli Institute for Theoretical Physics Conference in Santa Barbara, CA, May 2025
 - Oral presentation on JWST results for the K2-22b project
- NASA 2025 Exoplanet Explorers Program, (virtual) February 2025
 - Oral presentation on JWST results for the K2-22b project
- American Astronomical Society (AAS) Winter Meeting 245 in National Harbor, MD, January 2025
 - Oral presentation on JWST results for the K2-22b project
 - Press Briefing on JWST results for the K2-22b project
- The Second Assembly of the Order of the Octopus, August 2024
 - Oral presentation on TRAPPIST-1 PPOs with ATA project
- Escape from Exoplanets Workshop at Imperial College in London, UK, June 2024
 - Oral presentation on preliminary JWST results for the K2-22b project
- Exoplanets V in Leiden, Netherlands, June 2024
 - Poster presentation on preliminary JWST results for the K2-22b project
- American Astronomical Society (AAS) Summer Meeting 244 in Madison, WI, June 2024
 - Oral presentation on preliminary JWST results for the K2-22b project
- AbSciCon in Providence, Rhode Island, May 2024
- Extreme Solar Systems V in Christchurch, New Zealand, March 2024
 - Poster presentation on K2-22b with JWST project
- American Astronomical Society (AAS) Winter Meeting 243 in New Orleans, LA, January 2023
 - Oral presentation on TRAPPIST-1 PPOs with ATA project
- URSI GASS in Sapporo, Japan, August 2023
 - Oral Presentation to promote the PSETI Center
- PSETI Symposium at PSU, June 2023
 - Oral presentation on ongoing analysis of TRAPPIST-1 observations at ATA
- AbGradCon in San Diego, CA, May 2023
 - Poster presentation of the published SGL project
 - Proposal Writing Retreat (workshop before AbGradCon) on Catalina Island, CA
- SETI Workshop at Allen Telescope Array (ATA), April 2023

- Oral presentation on SETI pipeline progress at ATA
- American Astronomical Society (AAS) Winter Meeting 241 in Seattle, WA, Jan 2023
 - Oral presentation on published SGL project
- The First PSETI Symposium at Penn State University, June 2022
 - Oral presentation on the submitted SGL project
- The First Assembly of the Order of the Octopus (virtual), August 2020
 - Oral presentation on commensal radio SETI
- American Astronomical Society (AAS) Winter Meeting 235 in Honolulu, HI, January 2020
 - Poster presentation on undergraduate AGN outflow research
- American Astronomical Society (AAS) Summer Meeting 234 in St. Louis, MO, June 2019
 - Poster presentation on undergraduate AGN jet simulation research